

DBMS PROJECT

RESTAURANT MANAGEMENT SYSTEM

SRM University-AP
J.R V Manikanta





Project Overview

- A Restaurant Management System is software designed to manage restaurant operations efficiently, from order processing to inventory control.
- Purpose: Streamline order processing, billing, inventory management, and table reservations in a unified platform.
- Importance of DBMS: Enables efficient data storage, quick retrieval, and centralized management of all restaurant information.
- Relational Database Model: Provides structured data handling through organized tables with defined relationships, ensuring data integrity and consistency.

Project Background

- Manual systems are prone to errors and delays, causing operational bottlenecks
- Automation is essential to improve accuracy, speed, and workflow efficiency
- Key benefits include enhanced accuracy, faster order processing, and streamlined operations
- Real-world impact: competitive advantage and improved customer satisfaction



System Description

- General workflow from order placement to billing and inventory updates
- Key operations: order processing, billing, inventory management, reservations
- Database-driven system ensuring data integrity and availability
- Improves restaurant workflow and enhances customer experience



User Roles



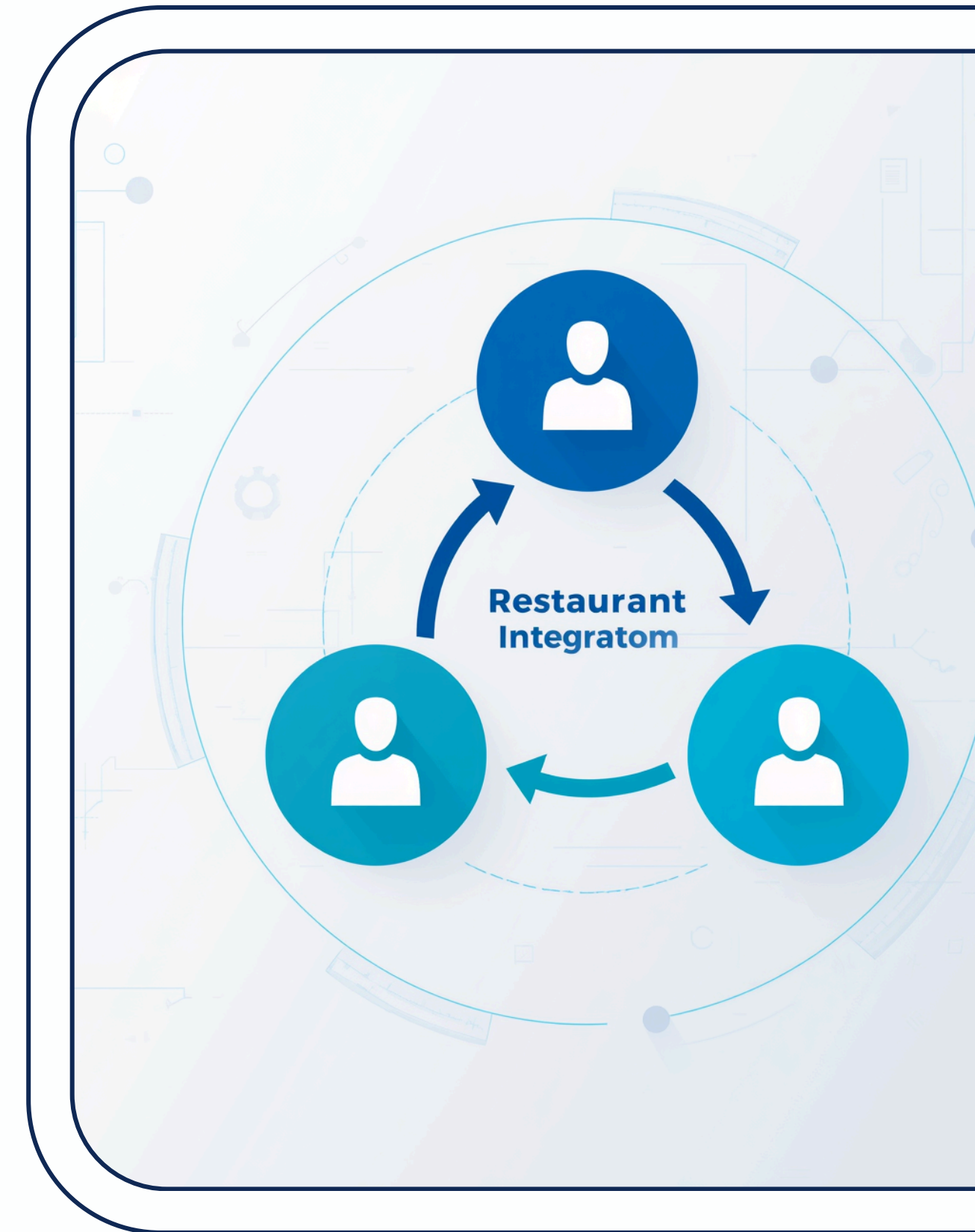
Admin

Full control over system settings, data management, user permissions, and complete access to all modules and analytics dashboards.

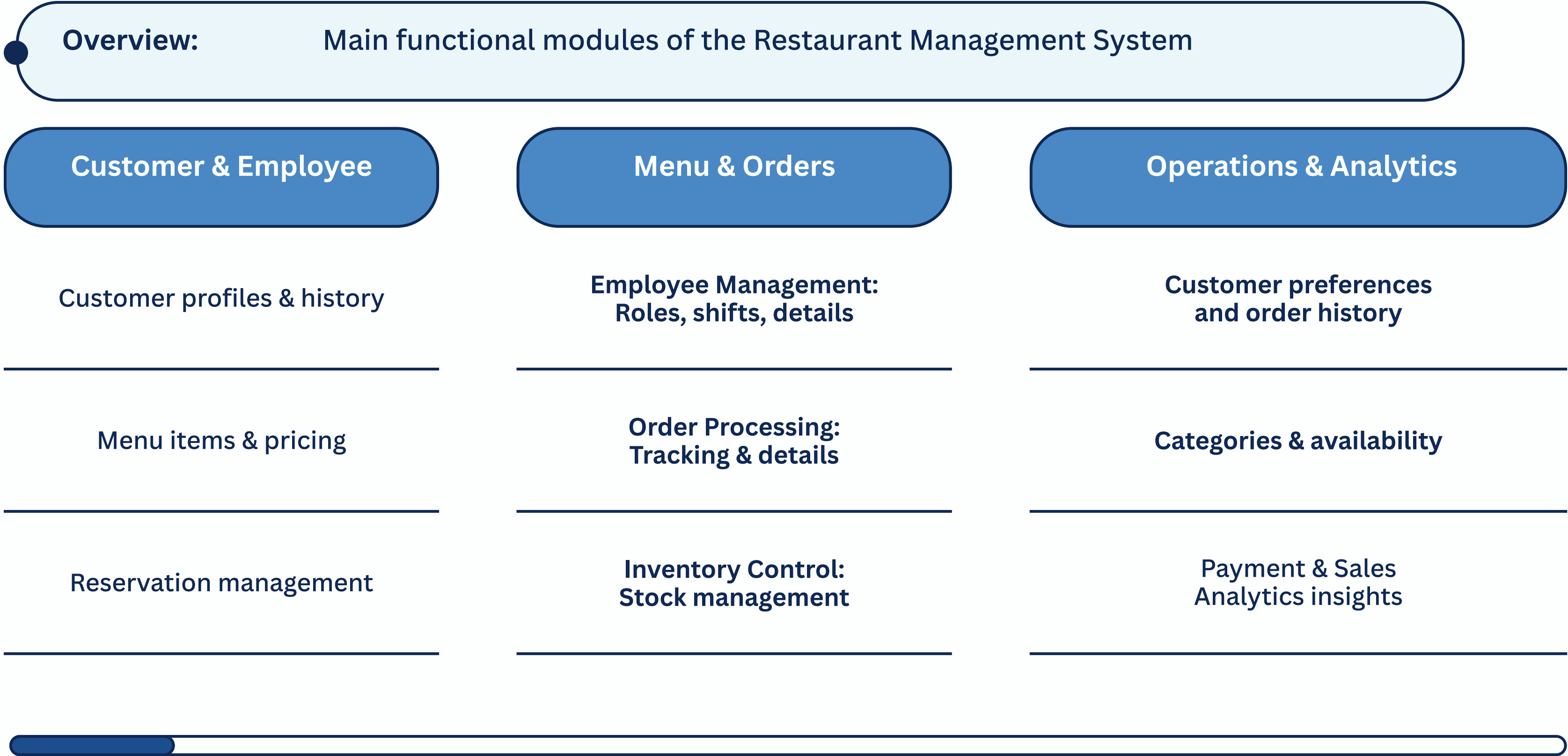


Staff

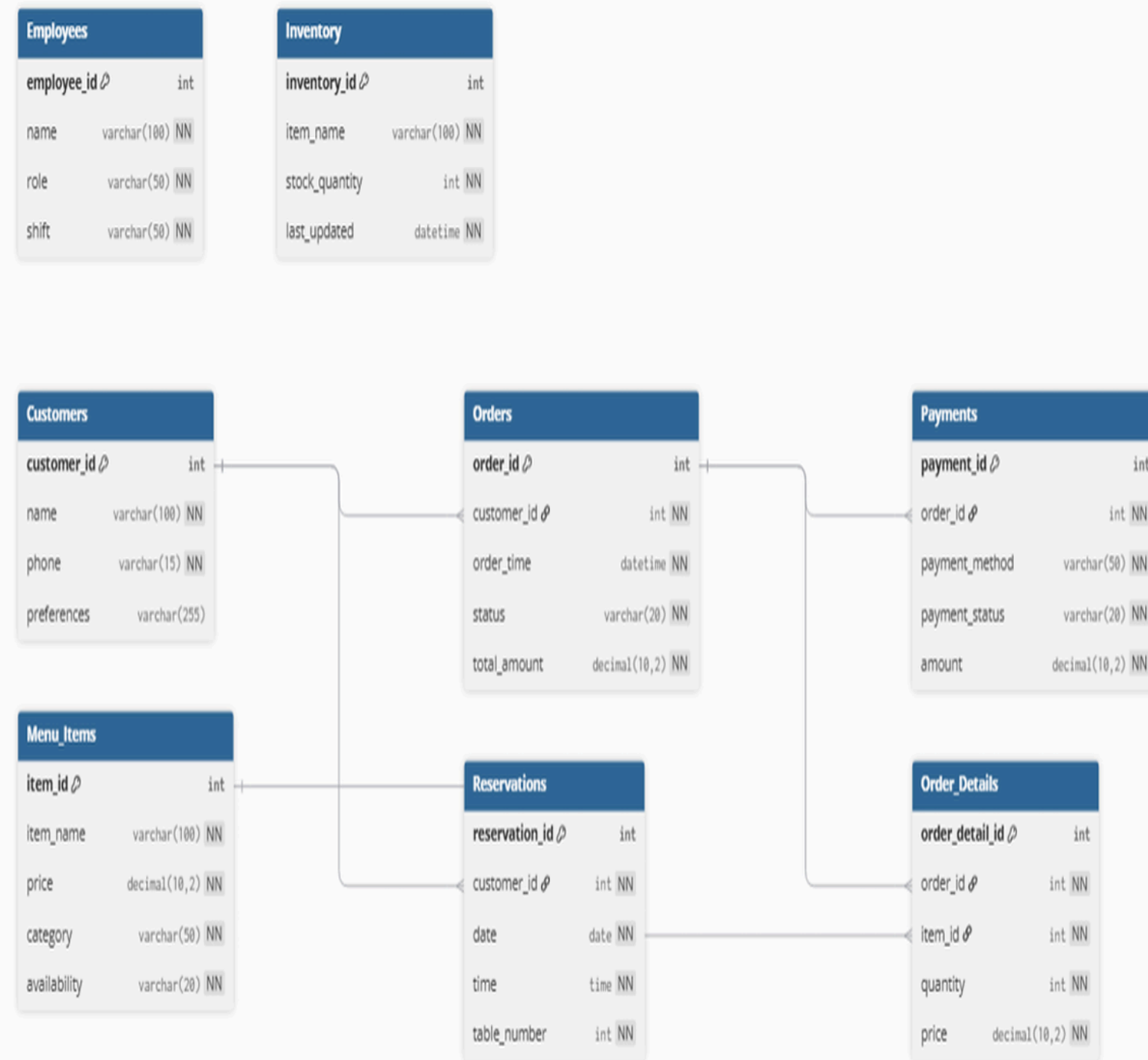
Manage daily operations including order processing, inventory updates, billing transactions, and customer service tasks.



Modules of System



The Entity-Relationship diagram illustrates the core database structure of our Restaurant Management System. Key entities include Customers, Employees, Menu_Items, Orders, Order_Details, Reservations, Inventory, and Payments. Critical relationships connect Orders to both Customers and Menu_Items, while Order_Details links individual items to their parent orders. Reservations reference both Customers and table availability, and Payments are tied to completed Orders. This relational structure ensures data integrity and enables efficient querying across all restaurant operations.



Primary Keys

CustomerID uniquely identifies each customer record

OrderID tracks individual orders placed in the system

MenuItemID identifies each dish in the menu catalog

PaymentID, ReservationID, EmployeeID for other entities

Foreign Keys

Orders.CustomerID references Customers table

Order_Details.OrderID links to Orders table

Payments.OrderID ensures billing integrity

Reservations.CustomerID maintains booking relationships

Key Concepts

JOINS combine data across related tables efficiently

Views simplify complex queries for reporting

Indexing speeds up data retrieval performance

Triggers automate actions on data changes

SQL Queries & Views

Purpose:

Database query examples and data access methods

JOINS

Combine customer & order data

SUM for totals calculation

Customer_Order_View

Aggregate Functions

INNER JOIN:
Link related tables

AVG Function:
Sales averages

Billing_View:
Simplified access

Views

Retrieve complete order
history with details

Revenue & performance metrics

Pre-defined queries for
common operations

THANK YOU

For queries:-

Email :- manikanta_jannu@srmap.edu.in

